28 January 2026, Shift 2

1. The time period of a simple harmonic oscillator is $T = 2\pi\sqrt{\frac{k}{m}}$. Measured value of mass (m) of the object is 10 g with an accuracy of 10 mg and time for 50 oscillations of the spring is found to be 60 s using a watch of 2 s resolution. Percentage error in determination of spring constant (k) is ——— %.

a. 6.76

b. 3.43

c. 3.35

d. 7.60

2. Identify the correct statements:

A. Electrostatic field lines form closed loops .

B. The electric field lines point radially outward when charge is greater than zero.

C. The Gauss-Law is valid only for inverse-square force.

D. The workdone in moving a charged particle in a static electric field around a closed path is zero.

E. The motion of a particle under Coulomb's force must take place in a plane.

Choose the correct answer from the options given below :

a. A, B, C, D Only

b. A, B, D, E Only

c. A, C, E Only

d. B, C, D, E Only

3. A particle starts moving from time $t = 0$ and its coordinate is given as $x(t) = 4t^3 - 3t$

A. The particle returns to its original position (origin) 0.866 units later

B. The particle is 1 unit away from origin at its turning point

C. Acceleration of the particle is non-negative

D. The particle is 0.5 units away from origin at its turning point

E. Particle never turns back as acceleration is non-negative

Choose the correct answer from the options given below :

a. A, B, C Only

b. C, E Only

c. A, C, D Only

d. A, C Only

4. The speed of a longitudinal wave in a metallic bar is 400 m/s. If the density and Young's modulus of the bar material are increased by 0.5% and 1%, respectively then the speed of the wave is changed approximately to_____ m/s.

a. 401

b. 399

c. 398

d. 402

5. Which one of the following is not a measurable quantity?

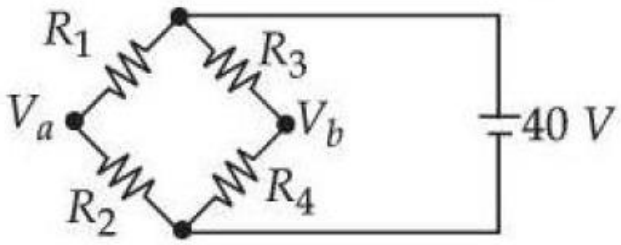
a. Resistance

b. Displacement current

c. Voltage

d. Voltage difference

6. A Wheatstone bridge is initially at room temperature and all arms of the bridge have same value of resistances $\left(R_1 = R_2 = R_3 = R_4\right)$. When R_3 resistance is heated to some temperature, its resistance value has gone up by 10%. The potential difference $\left(V_a - V_b\right)$ (after R_3 is heated) is ——— V.



a. 1.05

b. 0.95

c. 2

d. 0

7. The mean free path of a molecule of diameter 5×10^{-10} m at the temperature 41°C and pressure 1.38×10^{5} Pa, is given as —— m. (Given $k_B = 1.38\times10^{-23}$ J/K).

a. $10\sqrt{2}\times10^{-8}$

b. 2×10^{-8}

c. $2\sqrt{2}\times10^{-8}$

d. $2\sqrt{2}\times10^{-10}$

8. When the position vector $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ changes sign as $-\vec{r}$, which one of the following vector will not flip under sign change ?

a. Angular momentum

b. Velocity

c. Linear momentum

d. Acceleration

9. A plane electromagnetic wave is moving in free space with velocity $c = 3 \times 10^8$ m/s and its electric field is given as $\vec{E} = 54\sin(kz - \omega t)\hat{j}$ V/m, where $\hat{j}$ is the unit vector along y -axis. The magnetic field vector $\vec{B}$ of the wave is:

a. $+1.8\times10^{-7}\sin(kz-\omega t)\hat{i}\,\mathrm{T}$

b. $1.4\times10^{-7}\sin(kz-\omega t)\hat{i}\,\mathrm{T}$

c. $1.4\times10^{-7}\sin(kz-\omega t)\hat{k}\,\mathrm{T}$

d. $-1.8\times10^{-7}\sin(kz-\omega t)\hat{i}\,\mathrm{T}$

10. A nucleus has mass number α and radius R_α. Another nucleus has mass number β and radius R_β. If $\beta = 8\alpha$ then R_α / R_β is:

a. 0.5

b. 2

c. 1

d. 8

11. For a transparent prism, if the angle of minimum deviation is equal to its refracting angle, the refractive index n of the prism satisfies.

a. $\sqrt{2} < n < 2$

b. $1 < n < 2$

c. n≥2

d. $\sqrt{2} < n < 2\sqrt{2}$

12. A long cylindrical conductor with large cross section carries an electric current distributed uniformly over its cross-section. Magnetic field due to this current is :

A. maximum at either ends of the conductor and minimum at the midpoint

B. maximum at the axis of the conductor

C. minimum at the surface of the conductor

D. minimum at the axis of the conductor

E. same at all points in the cross-section of the conductor

Choose the correct answer from the options given below :

a. A, D Only

b. D Only

c. B, C Only

d. E Only

13. Identify the correct statements :

A. Effective capacitance of a series combination of capacitors is always smaller than the smallest capacitance of the capacitor in the combination.

B. When a dielectric medium is placed between the charged plates of a capacitor, displacement of charges cannot occur due to insulation property of dielectric.

C. Increasing of area of capacitor plate or decreasing of thickness of dielectric is an alternate method to increase the capacitance.

D. For a point charge, concentric spherical shells centered at the location of the charge are equipotential surfaces.

Choose the correct answer from the options given below :

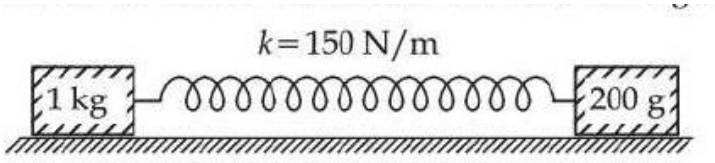
a. A, C and D Only

b. B and D Only

c. C and D Only

d. A, B and C Only

14. As shown in the figure, a spring is kept in a stretched position with some extension by holding the masses 1 kg and 0.2 kg with a separation more than spring natural length and are released. Assuming the horizontal surface to be frictionless, the angular frequency (in SI unit) of the system is :



a. 5

b. 27

c. 30

d. 20

15. Number of photons of equal energy emitted per second by a 6 mW laser source operating at 663 nm is ——— . (Given : $h = 6.63 \times 10^{-34}$ J.s and $c = 3 \times 10^{8}$ m / s)

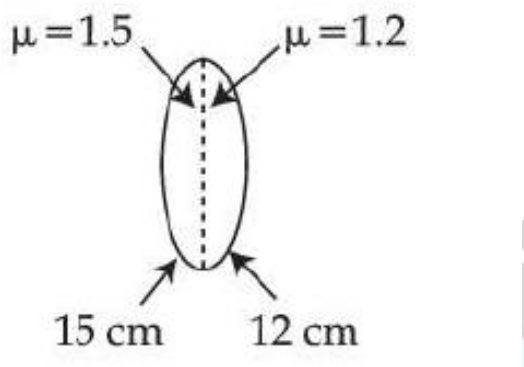
a. 10×10^{15}

b. 2×10^{16}

c. 5×10^{16}

d. 5×10^{15}

16. A biconvex lens is formed by using two thin planoconvex lenses, as shown in the figure. The refractive index and radius of curved surfaces are also mentioned in figure. When an object is placed on the left side of lens at a distance of 30 cm from the biconvex lens, the magnification of the image will be :



a. +2

b. -2

c. -2.5

d. +2.5

17. Match List - I with List - II.

List - I	**List - II**
A. Coefficient of viscosity	I. $\left[ML^{-1}T^{-2}\right]$
B. Surface tension	II. $\left[ML^{2}T^{-2}\right]$
C. Pressure	III. $\left[ML^{0}T^{-2}\right]$
D. Surface energy	IV. $\left[ML^{-1}T^{-1}\right]$

Choose the correct answer from the options given below :

a. A-I, B-III, C-II, D-IV

b. A-I, B-II, C-IV, D-III

c. A-IV, B-III, C-I, D-II

d. A-IV, B-I, C-II, D-III

18. In an experiment, a set of reading are obtained as follows $-1.24\,\text{mm}, 1.25\,\text{mm}, 1.23\,\text{mm}$, 1.21 mm . The expected least count of the instrument used in recording these readings is ——— mm.

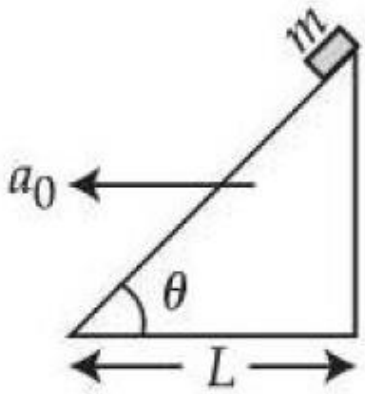
a. 0.01

b. 0.1

c. 0.001

d. 0.05

19. A small block of mass m slides down from the top of a frictionless inclined surface, while the inclined plane is moving towards left with constant acceleration a_0. The angle between the inclined plane and ground is θ and its base length is L. Assuming that initially the small block is at the top of the inclined plane, the time it takes to reach the lowest point of the inclined plane is_____



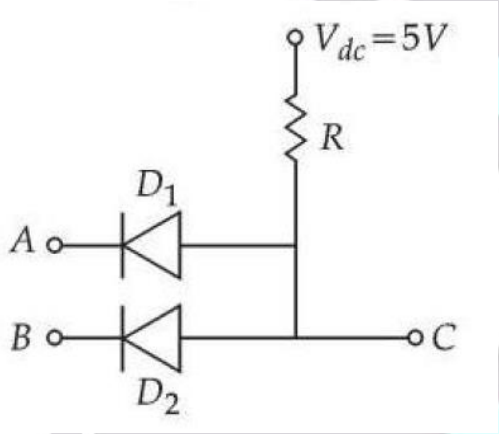
a. $\sqrt{\dfrac{2L}{g\sin\theta - a_0\cos\theta}}$

b. $\sqrt{\dfrac{4L}{g\sin 2\theta - a_0(1+\cos 2\theta)}}$

c. $\sqrt{\dfrac{2L}{g\sin 2\theta - a_0(1+\cos 2\theta)}}$

d. $\sqrt{\dfrac{4L}{g\cos^2\theta - a_0 \sin\theta\cos\theta}}$

20. Two p-n junction diodes D_1 and D_2 are connected as shown in figure. A and B are input signals and C is the output. The given circuit will function as a____ .



a. NAND Gate

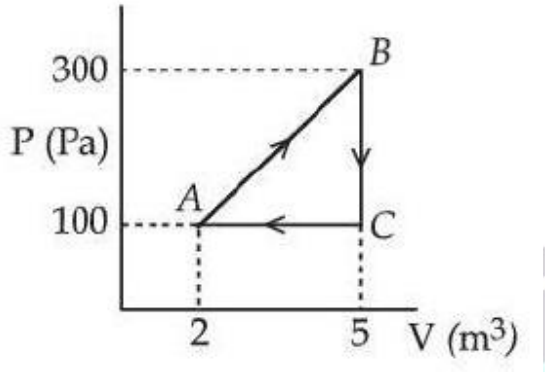
b. OR Gate

c. AND Gate

d. NOR Gate

21. Two tuning forks A and B are sounded together giving rise to 8 beats in 2 s. When fork A is loaded with wax, the beat frequency is reduced to 4 beats in 2 s . If the original frequency of tuning fork B is 380 Hz then original frequency of tuning fork A is ——— Hz.

22. A thermodynamic system is taken through the cyclic process ABC as shown in the figure. The total work done by the system during the cycle ABC is ——— J.



23. A fly wheel having mass 3 kg and radius 5 m is free to rotate about a horizontal axis. A string having negligible mass is wound around the wheel and the loose end of the string is connected to 3 kg mass. The mass is kept at rest initially and released. Kinetic energy of the wheel when the mass descends by 3 m is____ $\mathrm{J}.\left(\mathrm{g}=10\,\mathrm{m}/\mathrm{s}^2\right)$

24. An inductor stores 16 J of magnetic field energy and dissipates 32 W of thermal energy due to its resistance when an a.c. current of 2 A (rms) and frequency 50 Hz flows through it. The ratio of inductive reactance to its resistance is ____ . $(\pi=3.14)$

25. A beam of light consisting of wavelengths 650 nm and 550 nm illuminates the Young's double slits with separation of 2 mm such that the interference fringes are formed on a screen, placed at a distance of 1.2 m from the slits. The least distance of a point from the central maximum, where the bright fringes due to both the wavelengths coincide, is ____ $\times 10^{-5}\,\mathrm{m}$.

26. The correct increasing order of spin-only magnetic moment values of the complex ions $\left[\mathrm{MnBr}_4\right]^{2-}(\mathrm{A}),\left[\mathrm{Cu}\left(\mathrm{H}_2\mathrm{O}\right)_6\right]^{2+}(\mathrm{B}),\left[\mathrm{Ni}(\mathrm{CN})_4\right]^{2-}(\mathrm{C})$ and $\left[\mathrm{Ni}\left(\mathrm{H}_2\mathrm{O}\right)_6\right]^{2+}(\mathrm{D})$ is :

a. $\mathrm{A}=\mathrm{B}<\mathrm{D}<\mathrm{C}$

b. $C<\mathrm{B}<\mathrm{D}<\mathrm{A}$

c. $C = D < B < A$

d. $A = B < C < D$

27. A student has been given 0.314 g of an organic compound and asked to estimate Sulphur. During the experiment, the student has obtained 0.4813 g of barium sulphate. The percentage of sulphur present in the compound is ——— . (Given Molar mass in $gmol^{-1}\ S: 32, BaSO_4: 233$)

a. 42.10 %

b. 48.24 %

c. 21.05 %

d. 63.15 %

28. Total number of alkali insoluble solid sulphonamides obtained by reaction of given amines with Hinsberg's reagent is_____.

Aniline, N-Methylaniline, Methanamine, N, N-Dimethylmethanamine, N-Methyl methanamine, Phenylmethanamine, N-propylaniline, N-phenylaniline, N, N-Dimethylaniline, Allyl amine, Isopropyl amine

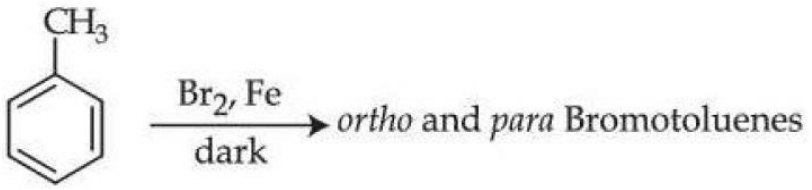
a. 2

b. 8

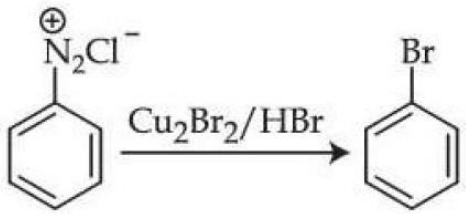
c. 5

d. 4

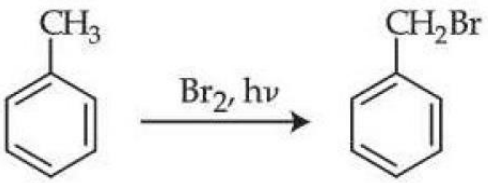
29. Which of the following reaction is NOT correctly represented ?

a. 

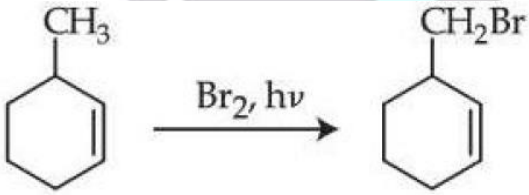
b.



c.



d.



30. For the given reaction;

$$CaCO_3 + 2HCl \longrightarrow CaCl_2 + H_2O + CO_2$$

If $90\,g CaCO_3$ is added to 300 mL of HCl which contains $38.55\% HCl$ by mass and has density $1.13\,g\,mL^{-1}$, then which of the following option is correct ?

Given molar mass of H, Cl, Ca and O are 1, 35.5, 40 and $16\,g\,mol^{-1}$ respectively.

a. 60.32 g of HCl remains unreacted

b. 64.97 g of HCl remains unreacted

c. 32.85 g of $CaCO_3$ remains unreacted

d. 97.30 g of HCl reacted

31. Given below are two statements :

Statement I : The increasing order of boiling point of hydrogen halides is $HCl < HBr < HI < HF$.

Statement II : The increasing order of melting point of hydrogen halides is $HCl < HBr < HF < HI$.

In the light of the above statements, choose the correct answer from the options given below:

a. Statement I is false but Statement II is true

b. Both Statement I and Statement II are false

c. Both Statement I and Statement II are true

d. Statement I is true but Statement II is false

32. Observe the following equilibrium in a 1 L flask.

$$A(g) \rightleftharpoons B(g)$$

At $T(K)$, the equilibrium concentrations of A and B are 0.5 M and 0.375 M respectively. 0.1 moles of A is added into the flask and heated to $T(K)$ to establish the equilibrium again. The new equilibrium concentrations (in M) of A and B are respectively

a. 0.557, 0.418

b. 0.367, 0.275

c. 0.742, 0.557

d. 0.53, 0.4

33. Consider the elements N, P, O, S, Cl and F . The number of valence electrons present in the elements with most and least metallic character from the above list is respectively.

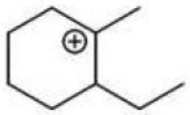
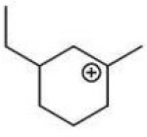
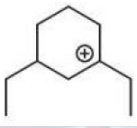
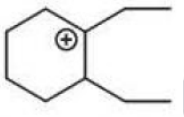
a. 6 and 7

b. 5 and 7

c. 7 and 5

d. 5 and 6

34. The cyclic cations having the same number of hyperconjugation are

(A)  (B)  (C)  (D) 

Choose the correct answer from the options given below :

a. A, C and D only

b. A and B Only

c. A and C Only

d. B and C Only

35. Identify the correct statements :

The presence of $-NO_2$ group in benzene ring

A. activates the ring towards electrophilic substitutions.

B. deactivates the ring towards electrophilic substitutions.

C. activates the ring towards nucleophilic substitutions.

D. deactivates the ring towards nucleophilic substitutions.

Choose the correct answer from the options given below :

a. B and C Only

b. B and D Only

c. C and A Only

d. A and D Only

36. The reactions which produce alcohol as the product are :

A. $CH_4 + O_2 \xrightarrow[\Delta]{Mo_2O_3}$

B. $2CH_3CH_3 + 3O_2 \xrightarrow[\Delta]{(CH_3COO)_2Mn}$

C. $(CH_3)_3CH \xrightarrow{KMnO_4}$

D. $2CH_4 + O_2 \xrightarrow{Cu/523\ K/100\ atm.}$

E. $CH_3 - CH = CH - CH_3 \xrightarrow{KMnO_4/H^+}$

Choose the correct answer from the options given below :

a. A, C and E Only

b. A and D Only

c. B, D and E Only

d. C and D Only

37. The plot of $\log_{10} K vs \frac{1}{T}$ gives a straight line. The intercept and slope respectively are (where K is equilibrium constant).

a. $-\frac{\Delta S^\circ R}{2 \cdot 303}, \frac{\Delta H^\circ R}{2 \cdot 303}$

b. $\frac{\Delta S^\circ}{2 \cdot 303R}, -\frac{\Delta H^\circ}{2 \cdot 303R}$

c. $-\frac{\Delta H^\circ}{2 \cdot 303R}, \frac{\Delta S^\circ}{2 \cdot 303R}$

d. $\frac{2 \cdot 303R}{\Delta H^\circ}, \frac{2 \cdot 303R}{\Delta S^\circ}$

38. Consider the following statements about manganate and permanganate ions. Identify the correct statements.

A. The geometry of both manganate and permanganate ions is tetrahedral.

B. The oxidation states of Mn in manganate and permanganate are +7 and +6 , respectively.

C. Oxidation of Mn (II) salt by peroxodisulphate gives manganate ion as the final product.

D. Manganate ion is paramagnetic and permanganate ions is diamagnetic.

E. Acidified permanganate ion reduces oxalate, nitrite and iodide ions.

Choose the correct answer from the options given below:

a. A, C and D Only

b. A and D Only

c. A, B and C Only

d. A, D and E Only

39. The wavelength of photon ' A ' is 400 nm . The frequency of photon ' B ' is $10^{16}\ s^{-1}$. The wave number of photon 'C' is $10^{4}\ cm^{-1}$. The correct order of energy of these photons is :

a. $C > B > A$

b. $B > A > C$

c. $A > B > C$

d. $A > C > B$

40. Consider the following aqueous solutions.

I. 2.2 g Glucose in 125 mL of solution.

II. 1.9 g Calcium chloride in 250 mL of solution.

III. 9.0 g Urea in 500 mL of solution.

IV. 20.5 g Aluminium sulphate in 750 mL of solution.

The correct increasing order of boiling point of these solutions will be :

[Given : Molar mass in $gmol^{-1}$: $H = 1, C = 12, N = 14, O = 16, Cl = 35.5, Ca = 40, Al = 27$ and $S = 32$]

a. II < III < I < IV

b. I < II < III < IV

c. III < I < II < IV

d. II < III < IV < I

41. Match List - I with List - II according to shape.

List - I		**List - II**	
A.	XeO_3	I.	BrF_5
B.	XeF_2	II.	NH_3
C.	XeO_2F_2	III.	$[I_3]^-$
D.	$XeOF_4$	IV.	SF_4

Choose the correct answer from the options given below :

a. A-II, B-III, C-IV, D-I

b. A-II, B-I, C-III, D-IV

c. A-II, B-III, C-I, D-IV

d. A-III, B-II, C-IV, D-I

42. Consider the following reactions

$$Na_2B_4O_7 \xrightarrow{\Delta} 2X + Y$$

$$CuSO_4 + Y \xrightarrow{\text{Non-Luminous flame}} Z + SO_3$$

$$2Z + 2X + \text{Carbon} \xrightarrow{\text{Luminous flame}} 2Q + Na_2B_4O_7 + CO$$

The oxidation states of Cu in Z and Q , respectively are :

a. +1 and +1

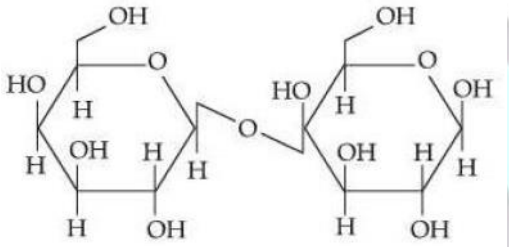
b. +1 and +2

c. +2 and +1

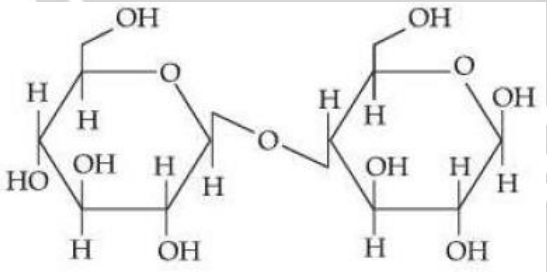
d. +2 and +2

43. Structures of four disaccharides are given below. Among the given disaccharides, the non-reducing sugar is :

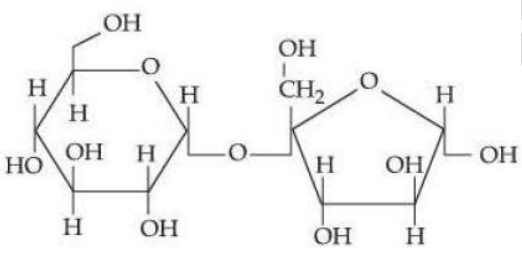
a.



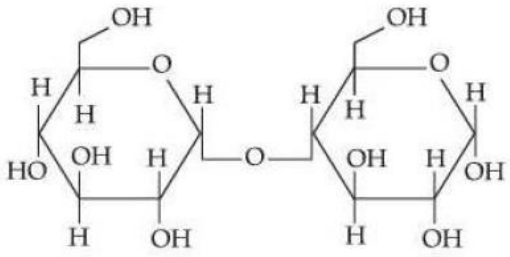
b.



c.



d.



44. A student performed analysis of aliphatic organic compound ' X ' which on analysis gave $C = 61.01\%$, $H = 15.25\%, N = 23.74\%$.

This compound, on treatment with HNO_2 / H_2O produced another compound 'Y' which did not contain any nitrogen atom. However, the compound ' Y ' upon controlled oxidation produced another compound 'Z' that responded to iodoform test.

The structure of ' X ' is :

a. $CH_3CH_2CH_2NH_2$

b. $CH_3-\underset{\underset{NH_2}{|}}{CH_2CH}-CH_3$

c. $\begin{matrix} CH_3 \\ CH_3 \end{matrix}\!\!\diagdown\!\!\diagup CH-NH_2$

d. $Ph-\underset{\underset{CH_3}{|}}{CH}-NH_2$

45. The correct order of acidic strength of the major products formed in the given reactions, is :

$$PhNH_2 \xrightarrow[(2)\ CuCN_2+HCl(<5^\circ C)]{(1)} [A]$$

A.

B. $CH_3CH_2CHO \xrightarrow[\Delta]{\left[Ag(NH_3)_2\right]^+, OH^-} [B]$

C. $CH_4 + O_2 \xrightarrow[(ii)\ Na_2Cr_2O_7/H^+]{(i)\ Mo_2O_3} [C]$

D. $PhCH_2MgBr + CO_2 \xrightarrow[H_3O^+]{\text{Dry ether}} [D]$

Choose the correct answer from the options given below :

a. $A > D > C > B$

b. $C > B > A > D$

c. $A > D > B > C$

d. $C > A > D > B$

46. The number of isoelectronic species among $Sc^{3+}, Cr^{2+}, Mn^{3+}, Co^{3+}$ and Fe^{3+} is 'n''. If 'n'' moles of AgCl is formed during the reaction of complex with formula $CoCl_3(en)_2NH_3$ with excess of $AgNO_3$ solution, then the number of electrons present in the t_{2g} orbital of the complex is ——— .

47. A volume of x mL of $5MNaHCO_3$ solution was mixed with 10 mL of $2MH_2CO_3$ solution to make an electrolytic buffer. If the same buffer was used in the following electrochemical cell to record a cell potential of 235.3 mV , then the value of $x =$ ——— mL (nearest integer).

Consider upto one place of decimal for intermediate calculations

Given :

$E^{o}_{HSnO_2^- | Sn(OH)_6^{2-}} = -0.9\,V$

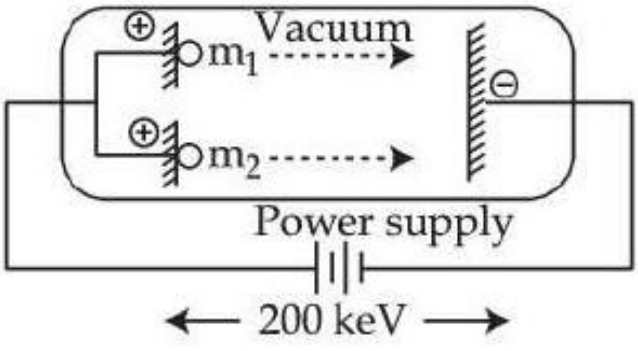
$E^{o}_{Bi_2O_3 | Bi} = -0.44\,V$

$pKa_{(H_2CO_3)} = 6.11$

$\frac{2.303\,RT}{F} = 0.059\,V$

$\text{Antilog}\,(1.29) = 19.5$

48. Two positively charged particles m_1 and m_2 have been accelerated across the same potential difference of 200 keV as shown below.



[Given mass of $m_1 = 1amu$ and $m_2 = 4amu$]

The deBroglie wavelength of m_1 will be x times of m_2. The value of x is —— (nearest integer)

49. For strong electrolyte Λ_m increases slowly with dilution and can be represented by the equation

$$\Lambda_m = \Lambda_m^\circ - Ac^{1/2}$$

Molar conductivity values of the solutions of strong electrolyte AB at $18^\circ C$ are given below :

$c\ [mol\ L^{-1}]$	0.04	0.09	0.16	0.25
$\Lambda_m\ [S\ cm^2\ mol^{-1}]$	96.1	95.7	95.3	94.9

The value of constant A based on the above data [in $Scm^2\ mol^{-1}/(mol/L)^{1/2}$] unit is ____ .

50. $A \longrightarrow B$ (first reaction) $C \longrightarrow D$ (second reaction) Consider the above two first-order reactions. The rate constant for first reaction at 500 K is double of the same at 300 K . At $500\,K, 50\%$ of the reaction becomes complete in 2 hour. The activation energy of the second reaction is half of that of first reaction. If the rate constant at 500 K of the second reaction becomes double of the rate constant of first reaction at the same temperature; then rate constant for the second reaction at 300 K is ____ $\times 10^{-1}$ hour $^{-1}$ (Nearest integer).

51. Let $[\cdot]$ denote the greatest integer function. Then $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \left(\frac{12(3+[x])}{3+[\sin x]+[\cos x]} \right) dx$ is equal to :

a. $11\pi + 2$

b. $12\pi + 5$

c. $13\pi + 1$

d. $15\pi + 4$

52. Let the arithmetic mean of $\frac{1}{a}$ and $\frac{1}{b}$ be $\frac{5}{16}, a > 2$. If α is such that $a, 4, \alpha, b$ are in A.P., then the equation $\alpha x^2 - ax + 2(\alpha - 2b) = 0$ has

a. complex roots of magnitude less than 2

b. one root in $(0, 2)$ and another in $(-4, -2)$

c. one root in (1,4) and another in ($-2, 0$)

d. both roots in the interval $(-2, 0)$

53. Considering the principal values of inverse trigonometric functions, the value of the expression $\tan\left(2\sin^{-1}\left(\frac{2}{\sqrt{13}}\right) - 2\cos^{-1}\left(\frac{3}{\sqrt{10}}\right)\right)$ is equal to :

a. $\frac{33}{56}$

b. $\frac{16}{63}$

c. $-\frac{16}{63}$

d. $-\frac{33}{56}$

54. An ellipse has its center at $(1, -2)$, one focus at $(3, -2)$ and one vertex at $(5, -2)$. Then the length of its latus rectum is :

a. 6

b. $4\sqrt{3}$

c. $\frac{16}{\sqrt{3}}$

d. $6\sqrt{3}$

55. Let the ellipse $E:\frac{x^2}{144}+\frac{y^2}{169}=1$ and the hyperbola $H:\frac{x^2}{16}-\frac{y^2}{\lambda^2}=-1$ have the same foci. If e and L respectively denote the eccentricity and the length of the latus rectum of H, then the value of $24(e+L)$ is :

a. 148

b. 126

c. 67

d. 296

56. The probability distribution of a random variable X is given below :

X	4k	$\frac{30}{7}$k	$\frac{32}{7}$k	$\frac{34}{7}$k	$\frac{36}{7}$k	$\frac{38}{7}$k	$\frac{40}{7}$k	6k
P(X)	$\frac{2}{15}$	$\frac{1}{15}$	$\frac{2}{15}$	$\frac{1}{5}$	$\frac{1}{15}$	$\frac{2}{15}$	$\frac{1}{5}$	$\frac{1}{15}$

If $E(X)=\frac{263}{15}$, then $P(X<20)$ is equal to :

a. $\frac{3}{5}$

b. $\frac{14}{15}$

c. $\frac{8}{15}$

d. $\frac{11}{15}$

57. Let the circle $x^2+y^2=4$ intersect x-axis at the points $A(a,0), a>0$ and $B(b,0)$. Let $P(2\cos\alpha, 2\sin\alpha)$, $0<\alpha<\frac{\pi}{2}$ and $Q(2\cos\beta, 2\sin\beta)$ be two points such that $(\alpha-\beta)=\frac{\pi}{2}$. Then the point of intersection of AQ and BP lies on:

a. $x^2+y^2-4x-4y=0$

b. $x^2+y^2-4y-4=0$

c. $x^2+y^2-4x-4y-4=0$

d. $x^2+y^2-4x-4=0$

58. The sum of all the elements in the range of

$$f(x)=\text{Sgn}(\sin x)+\text{Sgn}(\cos x)+\text{Sgn}(\tan x)+\text{Sgn}(\cot x), \quad x\neq\frac{n\pi}{2}, n\in Z,$$

where $\text{Sgn}(t)=\begin{cases}1, & \text{if } t>0\\ -1, & \text{if } t<0\end{cases}$, is :

a. 2

b. 4

c. –2

d. 0

59. $\frac{6}{3^{26}}+\frac{10\cdot 1}{3^{25}}+\frac{10\cdot 2}{3^{24}}+\frac{10\cdot 2^2}{3^{23}}+\ldots+\frac{10\cdot 2^{24}}{3}$ is equal to :

a. 2^{25}

b. 2^{26}

c. 3^{26}

d. 3^{25}

60. Let P be a point in the plane of the vectors $\overrightarrow{AB}=3\hat{i}+\hat{j}-\hat{k}$ and $\overrightarrow{AC}=\hat{i}-\hat{j}+3\hat{k}$ such that P is equidistant from the lines AB and AC. If $|\overrightarrow{AP}|=\frac{\sqrt{5}}{2}$, then the area of the triangle ABP is:

a. 2

b. $\frac{3}{2}$

c. $\frac{\sqrt{26}}{4}$

d. $\frac{\sqrt{30}}{4}$

61. Let $f(x)=\lim_{\theta\to 0}\left(\frac{\cos \pi x - x^{\left(\frac{2}{\theta}\right)}\sin(x-1)}{1+x^{\left(\frac{2}{\theta}\right)}(x-1)}\right), x\in \mathbf{R}$. Consider the following two statements :

(I) $f(x)$ is discontinous at $x=1$.

(II) $f(x)$ is continous at $x=-1$.

Then,

a. Only (II) is True

b. Only (I) is True

c. Both (I) and (II) are True

d. Neither (I) nor (II) is True

62. Let $P_1 : y=4x^2$ and $P_2 : y=x^2+27$ be two parabolas. If the area of the bounded region enclosed between P_1 and P_2 is six times the area of the bounded region enclosed between the line $y=\alpha x,\ \alpha>0$ and P_1, then α is equal to :

a. 6

b. 15

c. 12

d. 8

63. Let $y = y(x)$ be the solution of the differential equation $x\frac{dy}{dx} - y = x^2 \cot x, x \in (0, \pi)$. If $y\left(\frac{\pi}{2}\right) = \frac{\pi}{2}$, then $6y\left(\frac{\pi}{6}\right) - 8y\left(\frac{\pi}{4}\right)$ is equal to :

a. 3π

b. π

c. -3π

d. $-\pi$

64. Let $f(x) = \int \frac{dx}{x^{\left(\frac{2}{3}\right)} + 2x^{\left(\frac{1}{2}\right)}}$ be such that $f(0) = -26 + 24\log_e(2)$. If $f(1) = a + b\log_e(3)$, where $a, b \in \mathbf{Z}$, then $a + b$ is equal to :

a. –11

b. –26

c. –5

d. –18

65. The sum of the coefficients of x^{499} and x^{500} in $(1+x)^{1000} + x(1+x)^{999} + x^2(1+x)^{998} + \ldots + x^{1000}$ is :

a. $^{1002}C_{501}$

b. $^{1002}C_{500}$

c. $^{1001}C_{501}$

d. $^{1000}C_{501}$

66. Given below are two statements :

Statement I: The function $f:\mathbf{R}\to\mathbf{R}$ defined by $f(x)=\dfrac{x}{1+|x|}$ is one-one.

Statement II: The function $f:\mathbf{R}\to\mathbf{R}$ defined by $f(x)=\dfrac{x^2+4x-30}{x^2-8x+18}$ is many-one.

In the light of the above statements, choose the correct answer from the options given below :

a. Statement I is true but Statement II is false

b. Statement I is false but Statement II is true

c. Both Statement I and Statement II are true

d. Both Statement I and Statement II are false

67. Let

Then the max $\{|z_1-z_2|:z_1\in A$ and $z_2\in B\}$ is :

a. $\dfrac{15}{2}$

$\frac{17}{2}$

b.

c. 8

d. 9

68. Let A be the focus of the parabola $y^2 = 8x$. Let the line $y = mx + c$ intersect the parabola at two distinct points B and C. If the centroid of the triangle ABC is $\left(\frac{7}{3}, \frac{4}{3}\right)$, then $(BC)^2$ is equal to :

a. 32

b. 41

c. 80

d. 89

69. Let $Q(a, b, c)$ be the image of the point $P(3, 2, 1)$ in the line $\frac{x-1}{1} = \frac{y}{2} = \frac{z-1}{1}$. Then the distance of Q from the line $\frac{x-9}{3} = \frac{y-9}{2} = \frac{z-5}{-2}$ is

a. 6

b. 7

c. 8

d. 5

70. Given below are two statements :

Statement I: $25^{13} + 20^{13} + 8^{13} + 3^{13}$ is divisible by 7.

Statement II : The integral part of $(7+4\sqrt{3})^{25}$ is an odd number. In the light of the above statements, choose the correct answer from the options given below :

a. Both Statement I and Statement II are false

b. Statement I is false but Statement II is true

c. Statement I is true but Statement II is false

d. Both Statement I and Statement II are true

71. Three persons enter in a lift at the ground floor. The lift will go upto 10th floor. The number of ways, in which the three persons can exit the lift at three different floors, if the lift does not stop at first, second and third floors, is equal to _____.

72. Let f be a differentiable function satisfying $f(x)=1-2x+\int_0^x e^{(x-t)}f(t)dt, x\in \mathbf{R}$ and let $g(x)=\int_0^x (f(t)+2)^{15}(t-4)^6(t+12)^{17}dt, x\in \mathbf{R}$. If p and q are respectively the points of local minima and local maxima of g, then the value of $|p+q|$ is equal to ____ .

73. Let $A=\begin{bmatrix}3 & -4\\ 1 & -1\end{bmatrix}$ and B be two matrices such that $A^{100}=100B+I$. Then the sum of all the elements of B^{100} is ____

74. If the distance of the point $P(43,\alpha,\beta), \beta<0$, from the line $\vec{r}=4\hat{i}-\hat{k}+\mu(2\hat{i}+3\hat{k}), \mu\in \mathbf{R}$ along a line with direction ratios $3,-1,0$ is $13\sqrt{10}$, then $\alpha^2+\beta^2$ is equal to ____

75. If $\sum_{r=1}^{25}\left(\frac{r}{r^4+r^2+1}\right)=\frac{p}{q}$, where p and q are positive integers such that $\gcd(p,q)=1$, then $p+q$ is equal to ——— .

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Answer Key

Q No.	Answer	Q No.	Answer	Q No.	Answer	Q No.	Answer
1	a	21	384	41	a	61	d
2	d	22	300	42	c	62	c
3	a	23	30	43	c	63	d
4	a	24	314	44	c	64	a
5	c	25	429	45	d	65	b
6	b	26	b	46	6	66	c
7	c	27	c	47	78	67	b
8	a	28	d	48	2	68	c
9	d	29	d	49	4	69	b
10	a	30	b	50	5	70	d
11	b	31	c	51	a	71	210
12	b	32	a	52	c	72	9
13	a	33	b	53	a	73	0
14	c	34	c	54	a	74	170
15	b	35	a	55	d	75	976
16	b	36	d	56	d		
17	c	37	b	57	b		
18	a	38	b	58	a		
19	b	39	b	59	b		
20	c	40	b	60	d		